Experiment 10.2

Analysis of Skip Distance and Tropospheric Propagation using MATLAB

Aim

To analyse the skip distance of ionospheric propagation and the line-of-sight range of tropospheric propagation, and to compare their coverage.

Software Required

MATLAB (MATLAB Online)

Procedure / Algorithm

- Take ionosphere height h = 250 km and angle of incidence 10° to 70°; compute skip distance = $2h \cdot \tan(\theta)$.
- Take ht = 100 m, hr = 50 m and compute LOS distance = $4.12(\sqrt{ht} + \sqrt{hr})$.
- Compute the coverage ratio = skip distance / LOS distance and plot the results.

Output

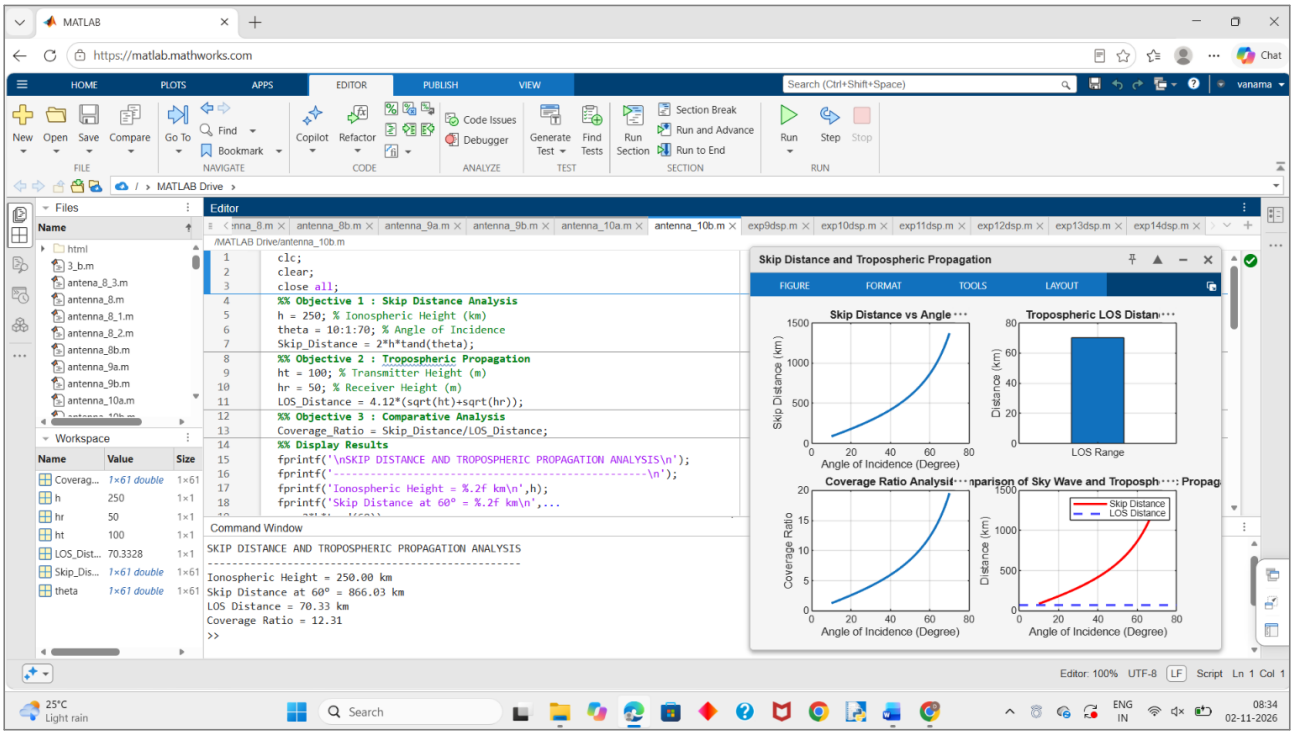


Fig. 10.2 Skip distance and tropospheric propagation analysis

Result

- Skip distance at 60° = 866.03 km; LOS distance = 70.33 km.
- Coverage ratio = 12.31, so ionospheric propagation covers a much larger distance than tropospheric LOS.